INFORMATICS INSTITUTE OF TECHNOLOGY

INDIVIDUAL CONTRIBUTION REPORT

3D Interactive Media Product

THE LOST ISLAND

Student name	K. Anuk Imira
Student ID / UOW number	20241663 | w2182300
Module	5MMCS007W – 3D Interactive Media
Group and assigned roles	Group 03 / Student 3 and Student 4; final project integration
Submission date	12 November 2026

Draft for personal review – replace bracketed notes and add your own screenshots before submission.

1. Introduction

The Lost Island is a LEGO-inspired 3D exploration game developed in Unity for Coursework 1. The project brings together a sequence of connected environments, beginning with an island exploration area and progressing towards ancient ruins and a hidden treasure location. The intention is to let the player explore the environment, follow a route through different areas and interact with objects that support the game's progression.

My assigned responsibilities covered Student 3 – Bridge and Character Movement, and Student 4 – Ancient Temple. These roles focused on the player's exploration route, the character controller and the environment and trigger-based interactions around the ancient temple. As the individual work was brought into the final group project, some elements were adjusted to fit the combined scene structure and the final gameplay direction. This report therefore explains both the responsibilities associated with my assigned roles and the work visible in the final project.

The project was developed using Unity 6 (6000.0.55f1) with the Universal Render Pipeline (URP). The final project uses separate scenes, including MainIsland, Cave, OpeningLobby and EndingLobby. The final presentation is not identical to the early individual task layouts because the group combined and adapted its work into one connected experience.

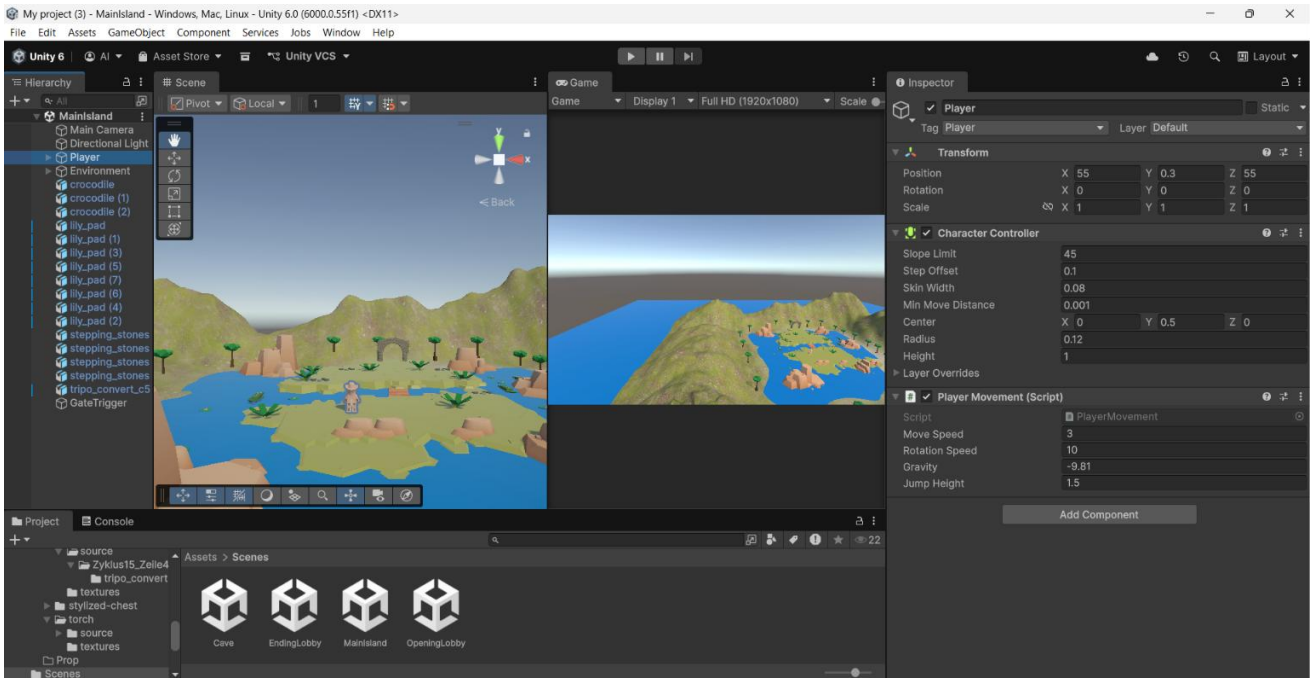
2. Individual Responsibilities

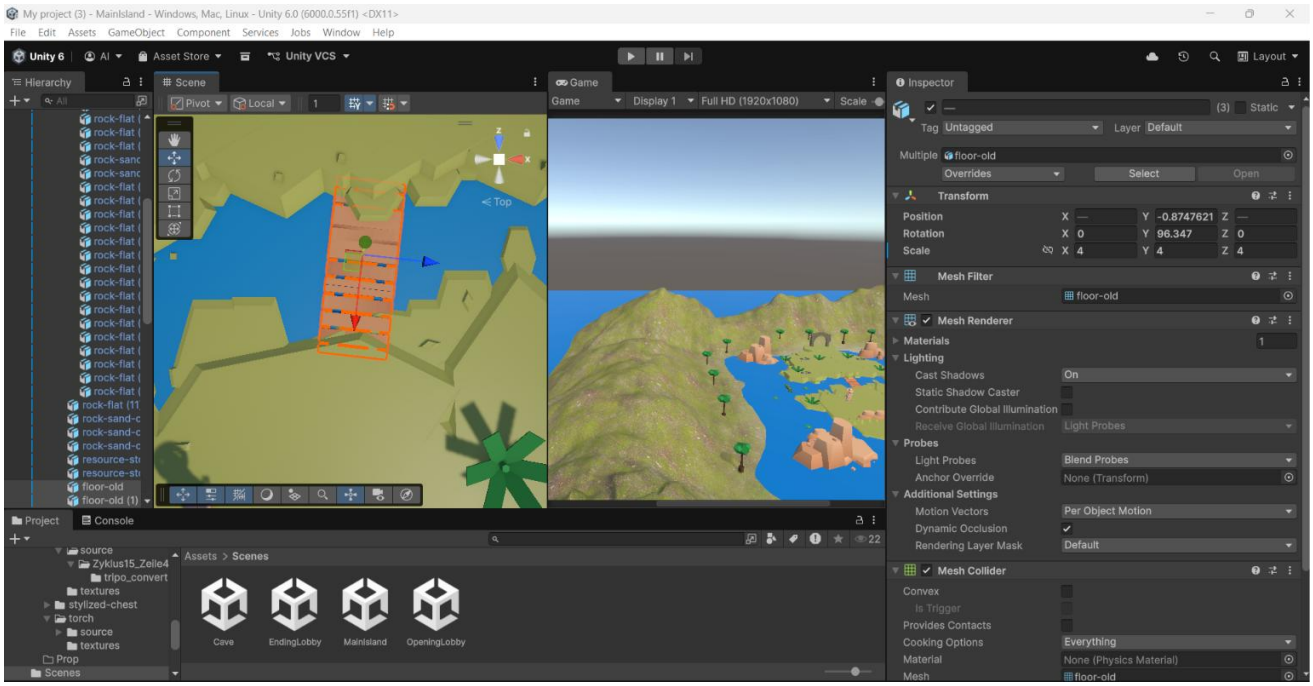
2.1 Student 3 – Bridge and character movement

The Student 3 task focused on creating the bridge and its surrounding environment, establishing a route for the character to explore, and testing the character controller. This role was important because the player needed to move through the island environment in a predictable way and reach the locations that form the game's progression.

For the final project, I worked on the player movement setup using a Player object with Unity's Character Controller component and a PlayerMovement script. The controller was configured to move the character across the ground while applying gravity. The movement system used keyboard input for navigation. I tested movement in the Unity Game view and adjusted the character and environment setup so that the player could move around the island rather than pass through solid environment pieces.

I also worked on the camera follow behaviour. A CameraFollow script was attached to the Main Camera and configured with the Player as its target. The follow offset and smoothing values were adjusted to keep the character visible while moving through the environment. This was part of making exploration easier to understand from the player's viewpoint.





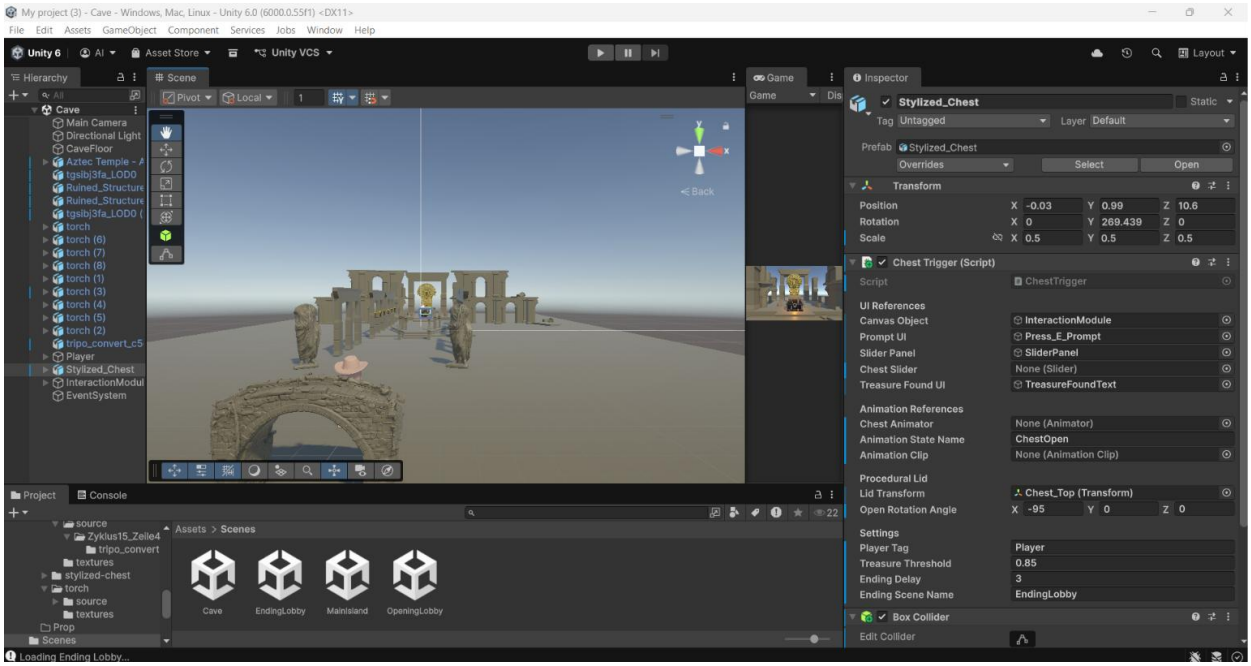
2.2 Student 4 – Ancient temple and trigger system

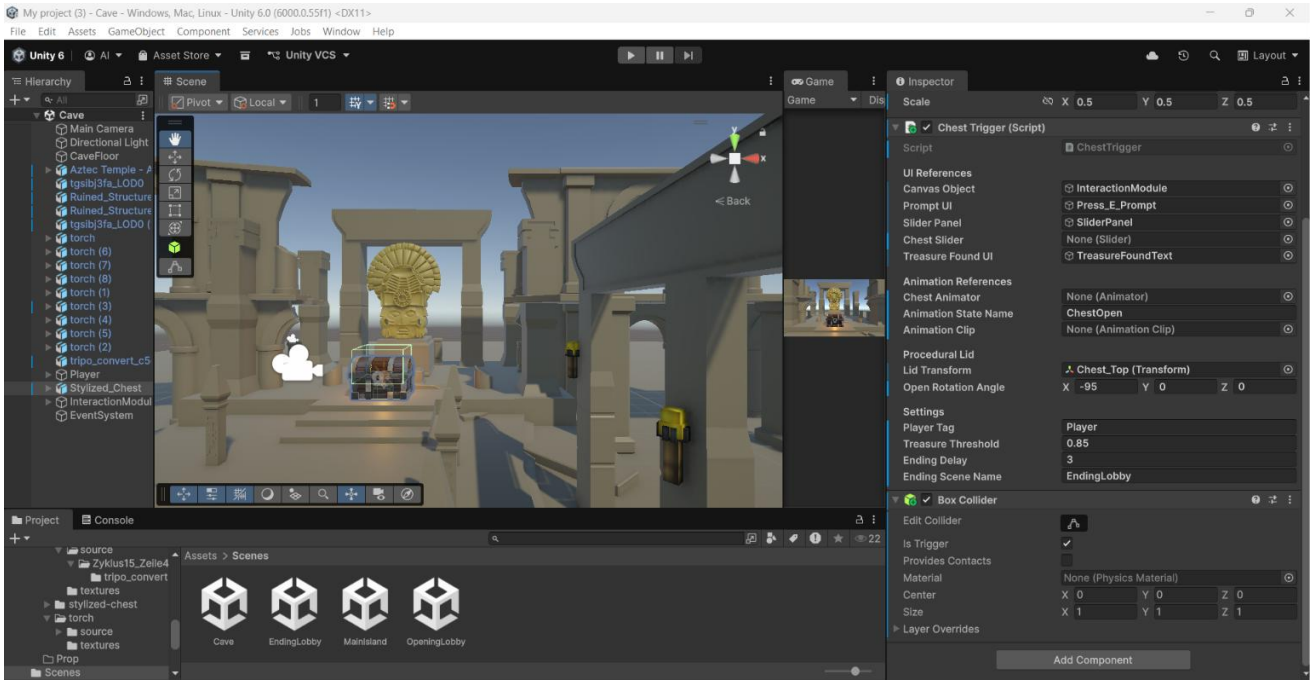
The Student 4 responsibility focused on the ancient temple entrance and interior, brick mechanisms and obstacles, and integration of a trigger system. The purpose of this area was to provide a point of interest in the island journey and to use interaction or trigger events to guide the player towards the next part of the experience.

In the combined project, the ancient environment and its connected locations were arranged as part of the exploration path. Solid environment pieces were checked for collision so that the player would not

pass through them. The Unity Terrain also provided its terrain collision. For imported land and structural meshes, Mesh Collider components were used where appropriate. This allowed the character controller to interact with the environment and follow the intended route.

The project also uses trigger-based interaction for progression and object interaction. In the final treasure area, the chest interaction uses an InteractionTrigger volume to detect when the player is close enough to interact. A prompt is displayed when the player enters the trigger area, and the player can use the E key to open the interaction interface. This is an example of how trigger-based interaction was carried into the final integrated experience. The chest is organised into separate parts, including Chest_Bottom and Chest_Top, so that the lid can be controlled independently.





3. Development and Technical Implementation

3.1 Player and environment collision

A key requirement for exploration was reliable collision between the player and the environment. The Player object uses a Character Controller rather than relying on a Rigidbody for character movement. Its capsule dimensions and centre were adjusted to fit the LEGO character. The terrain has a built-in Terrain Collider, while imported solid environment meshes require suitable colliders. I checked the land pieces and added Mesh Collider components where needed. This helped prevent the character from walking through solid ground and structural objects.

Collision testing also highlighted the difference between moving across a surface and climbing a steep surface. Where a slope is too steep for the Character Controller, the route needs a gentler ramp or a smoother transition. This is important when designing a bridge or path because the route must be both visually clear and traversable.

3.2 Character movement and camera

The PlayerMovement component handles keyboard movement and gravity through the Character Controller. The camera follows the player using a target Transform and a configurable offset. I adjusted the camera position and follow settings during testing to improve visibility of the player and the surrounding environment. The camera setup was also considered when preparing the cave scene, where a separate camera sequence was planned to introduce the treasure chest before following the player.

[Insert a screenshot of the PlayerMovement component and the CameraFollow component with their assigned references and values.]

3.3 Trigger-based interaction and treasure sequence

The final project includes an interactive treasure chest in the Cave scene. The chest model is separated into a base, lid and coin components. An interaction trigger is used to detect player proximity. When the player enters the trigger area, the interface can display a prompt. Pressing E opens the slider panel, allowing the player to control the lid position. The lid movement can be driven by an Animator or by a procedural rotation fallback, depending on the configured references.

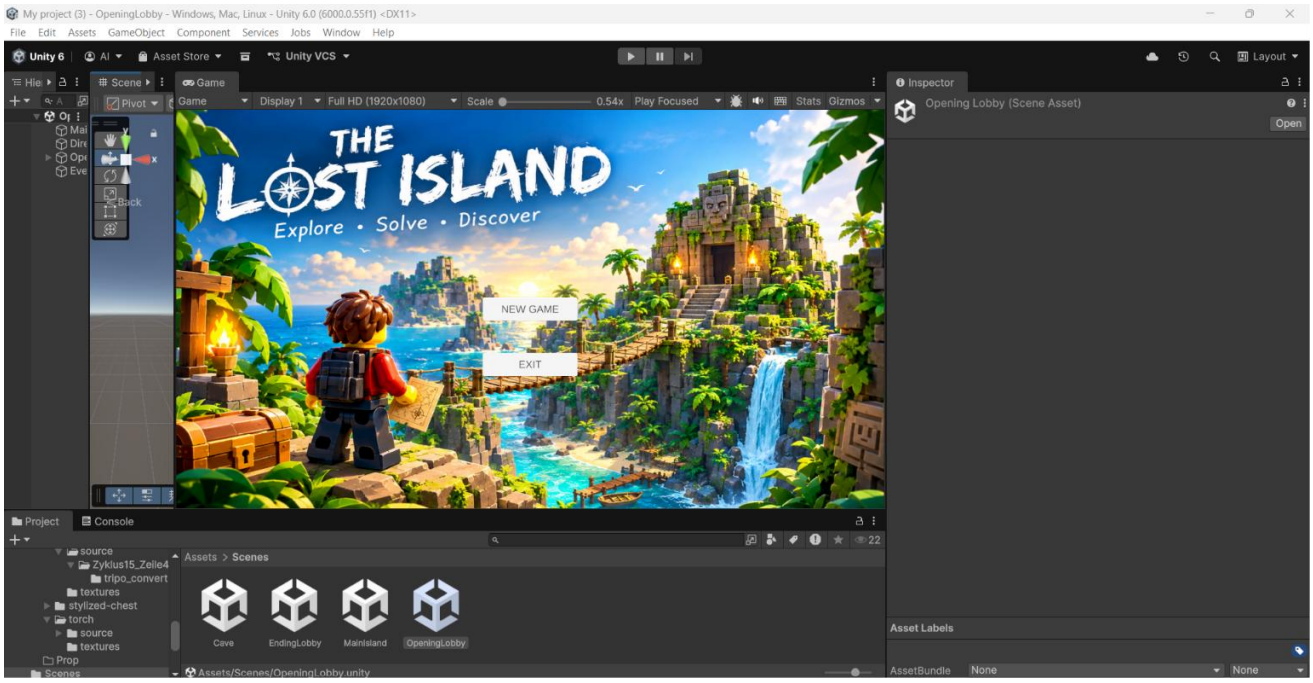
The user interface was built with a Canvas, a prompt text object and a slider panel containing a UI Slider. The player movement component is paused while the slider is being used so that keyboard movement does not interfere with the interaction. This final interaction demonstrates how a trigger, UI event and 3D object transform can work together.

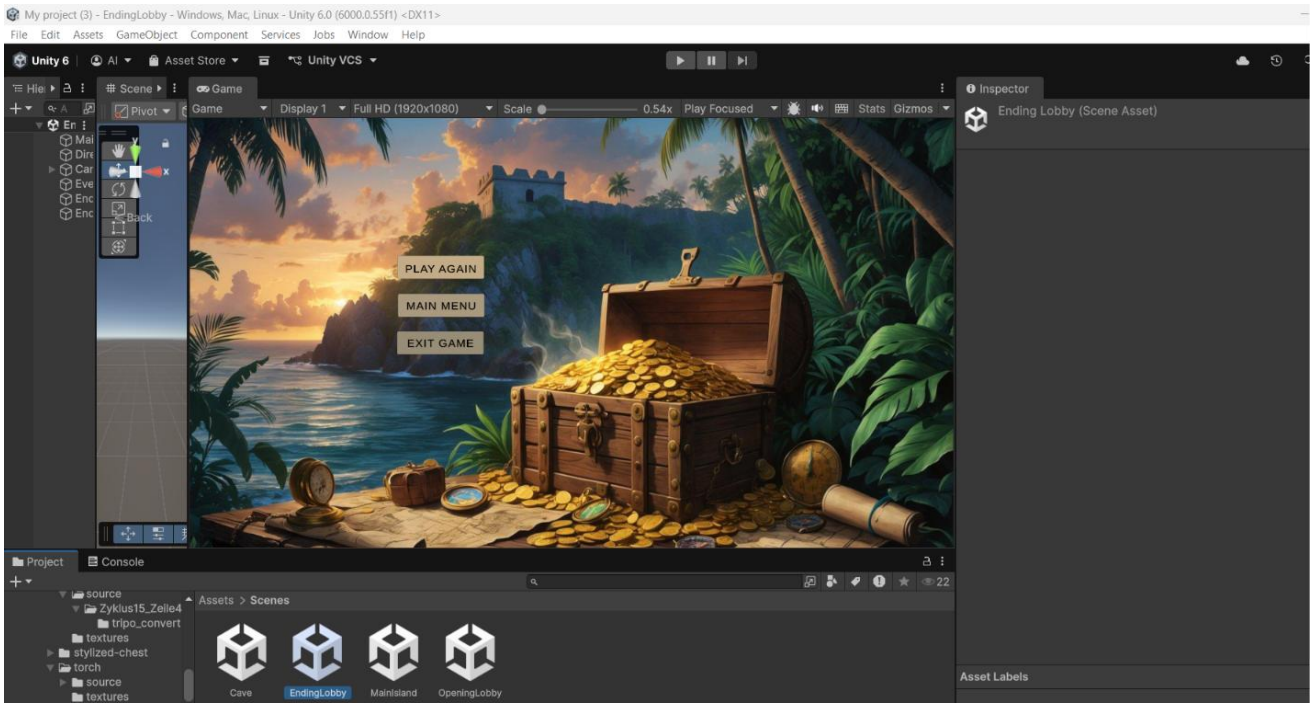
[Insert screenshots of the chest hierarchy, the InteractionTrigger Inspector, the prompt/slider in Game view, and the chest in its open state. Only include screenshots that show the final working version.]

3.4 Scene and lobby integration

The final project was organised into separate scenes to make the game flow easier to manage. The OpeningLobby scene provides the starting interface. The New Game button loads MainIsland. The Cave scene contains the treasure area, and the EndingLobby provides options after gameplay. The Play Again button reloads MainIsland, while the Main Menu button returns to OpeningLobby. These button events use Unity's SceneManager.LoadScene method through the OpeningLobbyButtons script.

The scenes were added to the Unity 6 Build Profiles Scene List so that they could be loaded by name at runtime. This was an important integration step because a scene can exist in the Project window but still be unavailable to SceneManager if it has not been added to the build scene list.


THE LOST ISLAND
Explore • Solve • Discover
NEW GAME
EXIT
Opening Lobby (Scene Asset)
Open
Cave
EndingLobby
MainIsland
OpeningLobby
Asset Labels


PLAY AGAIN
MAIN MENU
EXIT GAME
Ending Lobby (Scene Asset)
Cave
EndingLobby
MainIsland
OpeningLobby
Asset Labels

4. Challenges and Solutions

One challenge was making the character interact correctly with imported environment pieces. Objects that look like solid ground do not automatically block the player unless they have appropriate collision components. I addressed this by checking the colliders on the terrain and imported meshes and testing movement in Play mode.

Another challenge was setting up the camera so that it showed the character and environment clearly. A camera positioned too high, too low or too far away could show mostly sky or an unhelpful side view. I adjusted the follow offset and target reference and tested the result while moving the character.

The chest interaction also required careful coordination between the trigger volume, the UI and player movement. The prompt and slider needed to appear at the correct time, and the cursor needed to be available for slider interaction. The lid's original orientation also had to be preserved so that the slider did not rotate it into an incorrect position. I tested the interaction repeatedly and adjusted the object references and rotation behaviour.

Finally, scene navigation required the correct scene names and build configuration. I connected the lobby buttons to the appropriate public methods and added the scenes to Build Profiles. I then tested the New Game, Play Again and Main Menu paths.

5. Final Integration and Testing

The final group project differs from the separate responsibilities because the environments and systems were combined and adapted to form one experience. My contribution should therefore be understood in two stages: the work associated with the Student 3 and Student 4 responsibilities, and the subsequent work to make the combined project playable and navigable.

The following checks were used during development: the player could move using the keyboard; the Character Controller collided with the terrain and solid environment pieces; the camera followed the player; the player could reach the relevant environment areas; the chest prompt appeared when the player entered its interaction area; the slider controlled the chest lid; and the lobby buttons loaded their intended scenes.

[Before submission, test every item above in the final project and change this paragraph if any feature is incomplete. Add a small test table with Pass/Needs work and evidence screenshots.]

6. Reflection

Working on character movement and the ancient environment helped me understand how several Unity systems must work together to create a usable exploration experience. A character controller alone is not

enough: the environment needs suitable collision, the camera must present the route clearly, and trigger volumes must be placed and configured so that interactions occur at the right time.

I also gained experience integrating separate scenes and connecting UI buttons to scene-loading methods. Testing revealed that small configuration details, such as an unassigned target, an incorrect collider setting or a scene missing from Build Profiles, can prevent an otherwise correct system from working. Repeated testing and checking the Inspector were therefore important parts of my workflow.

The final project also demonstrated the importance of collaboration. Individual areas and systems needed to be adjusted to match the shared project rather than remain isolated demonstrations. This improved my understanding of project organisation, scene integration and the need to test the complete game flow after combining team contributions.

7. AI Code Assistance Statement

I used ChatGPT as a learning and troubleshooting aid while working in Unity. The guidance helped me understand Unity editor workflows, configure components, organise scene objects, troubleshoot character movement and camera behaviour, and connect UI buttons to scene navigation. I also used AI guidance to understand and adapt C# scripts for movement, camera following and interaction.

I tested the suggested settings and code in the Unity project and checked the results in the Editor. The final report should include the AI-use declaration required by the coursework brief, together with the relevant conversation transcripts and an accurate summary of how AI assistance was used. I should only claim authorship of code and systems that I personally implemented, modified or tested, and should explain any AI-generated or AI-assisted sections transparently.

8. Evidence to Add Before Submission

- Final MainIsland overview showing the exploration route and bridge area.
- Player Inspector showing Character Controller and PlayerMovement.
- Camera Inspector showing CameraFollow and its Player target.
- Ancient temple/ruins area in the final project.
- Trigger collider Inspector and a screenshot showing the interaction working.
- Cave scene showing the treasure chest and its hierarchy.
- Chest prompt and slider visible in the Game view.
- EndingLobby and OpeningLobby button layouts.
- Build Profiles Scene List with all required scenes enabled.
- A short test table confirming which features work in the final build.

Important drafting note: This is a first draft based on the project details available in the conversation and the uploaded example reports. Confirm the exact bridge, temple and trigger work you personally completed, insert your own Unity screenshots, and replace all bracketed instructions before submission. Do not leave any feature described as complete unless you have tested it in the final project.